

CULIACAN, Mexico

WMO: 764120

Lat: 24.6350N

Lon: 107.4411W

Elev: 39

StdP: 100.86

Time Zone: -7.00 (MEC)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.6	10.9	4.3	5.2	18.7	6.3	5.9	19.3	8.0	23.5	6.1	25.8	0.6	220	0.338

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	38.0	26.1	37.1	25.9	36.2	25.7	28.8	34.4	28.2	33.9	27.8	33.4	3.8	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
27.5	23.5	30.9	27.0	22.7	30.6	26.2	21.7	30.2	93.6	34.3	90.7	33.8	88.8	33.7	33.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.8	6.0	5.3	7.2	40.6	1.6	0.5	6.0	40.9	5.1	41.2	4.2	41.5	3.1	41.9
			6.0	30.6	1.4	1.2	4.9	31.5	4.1	32.2	3.3	32.8	2.3	33.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	25.9	20.2	20.9	22.3	24.2	27.1	30.2	30.7	30.3	30.0	29.1	24.9	21.2
	DBStd	4.33	2.37	1.94	1.79	1.49	1.74	1.57	1.35	1.37	1.40	1.80	2.29	2.17
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	17	11	2	1	0	0	0	0	0	0	0	0	3
	CDD10.0	5820	316	304	382	427	529	606	643	628	599	593	448	348
	CDD18.3	2796	68	73	124	177	271	356	384	369	349	334	198	93
	CDH23.3	33444	751	811	1455	2057	3270	4544	5036	4656	4285	3847	1873	860
(13)	CDH26.7	16993	226	267	622	1047	1847	2449	2759	2417	2157	2017	911	276
(14)	Wind	WSAvg	1.7	1.3	1.5	1.7	2.0	2.2	2.0	1.7	1.6	1.5	1.3	1.4
(15) Precipitation	PrecAvg	706	19	13	3	3	1	30	173	190	155	51	19	20
	PrecMax	1043	223	140	26	69	8	122	368	318	318	203	69	131
	PrecMin	419	0	0	0	0	0	0	61	49	63	0	0	0
	PrecStd	170	46	28	6	13	2	33	64	61	64	52	24	28
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.1	32.9	35.0	36.8	38.2	38.8	39.1	38.2	37.8	38.2	36.2	32.1
		MCWB	20.4	20.8	20.9	21.7	23.3	26.2	26.9	27.7	28.0	25.6	22.3	20.6
	2%	DB	30.2	31.1	32.9	34.8	37.0	37.2	37.2	37.0	36.4	37.0	34.8	30.9
		MCWB	19.6	20.4	20.6	21.1	23.2	25.8	26.6	27.5	27.5	25.2	22.4	20.2
	5%	DB	29.1	29.9	31.8	33.2	35.6	36.2	36.8	35.9	35.2	35.9	33.2	29.8
		MCWB	19.5	19.9	20.5	20.8	22.8	25.5	26.5	27.1	27.1	25.6	22.5	20.1
	10%	DB	27.9	28.2	30.2	32.1	34.2	35.2	35.5	34.9	34.2	34.8	32.1	28.1
		MCWB	18.9	19.4	20.1	20.6	22.5	25.2	26.2	27.1	27.0	25.4	22.3	19.6
	0.4%	WB	22.2	24.0	24.5	24.4	27.0	28.4	29.1	29.5	29.9	29.2	26.1	23.9
		MCDB	28.8	29.3	30.7	32.6	35.5	35.9	35.2	35.4	35.5	34.4	32.6	29.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	21.1	22.0	23.0	23.2	25.2	27.2	28.1	28.6	28.7	28.0	24.8	22.2
		MCDB	28.2	29.0	30.0	31.8	33.5	35.1	34.7	34.1	34.1	33.2	32.2	28.0
	5%	WB	20.4	21.0	21.7	22.3	24.6	26.8	27.2	28.1	28.1	27.1	23.9	21.1
		MCDB	27.4	27.7	29.3	30.7	32.7	34.4	33.8	33.5	33.5	32.4	30.8	27.0
	10%	WB	19.7	20.1	20.8	21.6	23.8	26.0	26.9	27.5	27.5	26.6	23.0	20.3
		MCDB	26.5	27.1	28.7	29.5	31.6	33.0	33.4	32.8	32.4	31.8	29.5	26.3
	5% DB	MDBR	15.7	15.6	17.2	17.1	15.7	11.1	10.7	10.2	10.0	12.5	14.7	14.9
		MCDBR	17.1	17.7	19.0	18.8	17.4	12.2	11.8	11.3	11.0	13.4	15.7	16.4
		MCWBR	8.4	8.9	9.0	8.1	6.9	4.8	4.4	4.3	4.4	5.2	6.6	8.1
		MCWBR	15.3	15.8	16.5	16.6	14.7	11.4	10.9	10.5	10.3	11.5	14.5	14.5
(40) Clear-Sky Solar Irradiance	taub	0.301	0.304	0.307	0.337	0.374	0.419	0.425	0.406	0.409	0.350	0.312	0.301	
		2.647	2.607	2.568	2.466	2.403	2.360	2.400	2.460	2.417	2.616	2.681	2.650	
	Ebn at Noon	970	987	993	961	918	870	865	884	877	924	950	956	
		82	92	100	113	121	125	120	113	115	90	79	79	
(44) All-Sky Solar Radiation	RadAvg	4.29	5.23	6.47	7.35	7.76	7.21	6.45	6.15	5.56	5.43	4.68	4.09	
	RadStd	0.34	0.41	0.23	0.38	0.32	0.31	0.29	0.21	0.23	0.27	0.25	0.20	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page